

21 A scientist used a sharp knife to cut a cube from a potato.

The cube had a mass of 3.10 g.

The scientist placed the cube in sucrose solution and measured the mass of the cube every 30 minutes for 120 minutes. The results are shown in the table.

time / minutes	mass of cube / g
0	3.10
30	3.31
60	3.40
90	3.43
120	3.43

After 120 minutes, the scientist measured:

- the turgor pressure of the potato cells
- the water potential of the potato cells
- the water potential of the sucrose solution.

Turgor pressure is the hydrostatic pressure within a cell that pushes the cell surface membrane against the cell wall.

Which values of potato cell turgor pressure, potato cell water potential and sucrose solution water potential after 120 minutes are consistent with the results?

	potato cell turgor pressure / MPa	potato cell water potential / MPa	sucrose solution water potential / MPa
A	0.0	−1.7	−1.2
B	1.2	−0.6	−0.6
C	0.7	−0.5	−1.2
D	0.7	−1.2	−0.5